

Name	Class	Index No.
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RAFFLES JUNIOR COLLEGE

Preliminary Examination

H1 BIOLOGY

8875/2

PAPER 2

9th September 2008

2 hours

Additional materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Write your answers **ONLY in the spaces** provided on the question paper.

Section B

Answer any **ONE** question.

Write your answer on the separate answer paper provided.

All workings for numerical answers must be shown.

At the end of the examination, **hand in your essay SEPARATELY.**

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	X
1	
2	
3	
4	
Section B	
Total	

This document consists of **10** printed pages.



Raffles Junior College
Internal Examination

[Turn over

Section A

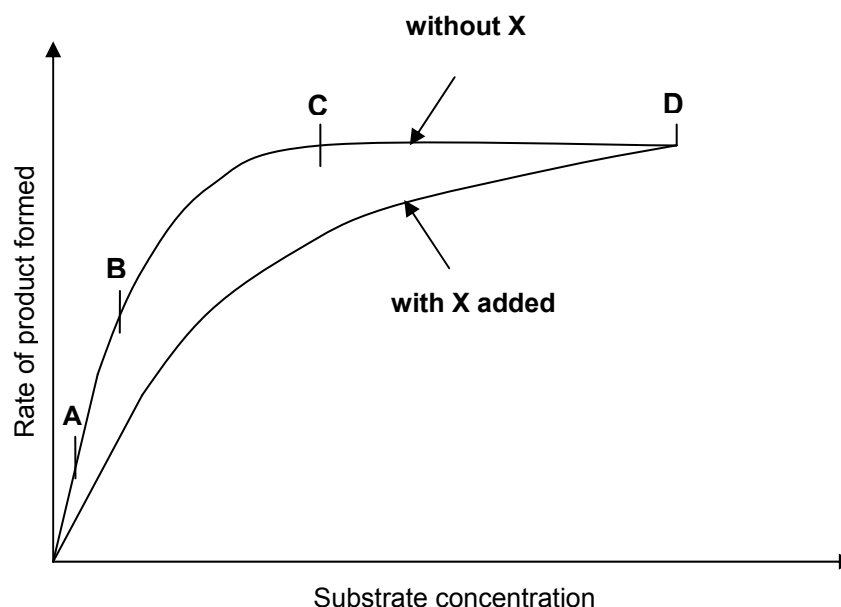
Answer **all** the questions in this section.

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Question 1

- (a) A student carried out an investigation to determine the effect of different concentrations of a substrate on the activity of an enzyme. Another set of experiments was carried out under the same conditions but with a compound **X** added. The results for both experiments were sketched out by the student in Fig.1.1 below.

Fig. 1.1



Account for the shape of the graph (without **X**),

(i) from **A** to **B**, and

- The activity of the enzyme is limited by substrate availability (wtte);
- As substrate concentration increases, increasing frequency of successful collisions (hence formation of enzyme-substrate complexes);
- leading to formation of more product per unit time; [max 2]

(ii) from **C** to **D**.

- Increasing substrate concentration does not increase the activity of the enzyme;
- Enzyme active sites saturated by substrate molecules;
- Concentration of enzyme is most probably limiting; [max 2]

(iii) Account for the effect of **X** on the enzyme.

- Enzyme undergoes competitive inhibition;
- **X** may be structurally similar to substrate;
- **X** competes with substrate for the occupation of active sites of enzyme molecules, reduced level of product formed (reduced level of enzyme activity);
- At higher substrate concentrations of substrate, substrate molecules out-compete inhibitor molecules for enzyme sites, hence inhibition can be overcome; [max 3]

- (b) When tadpoles turn into frogs, their tails become shorter and finally disappear. Fig. 1.2 shows how the concentration of a protein-digesting enzyme, cathepsin, found in lysosomes changes as a tadpole matures into a frog.

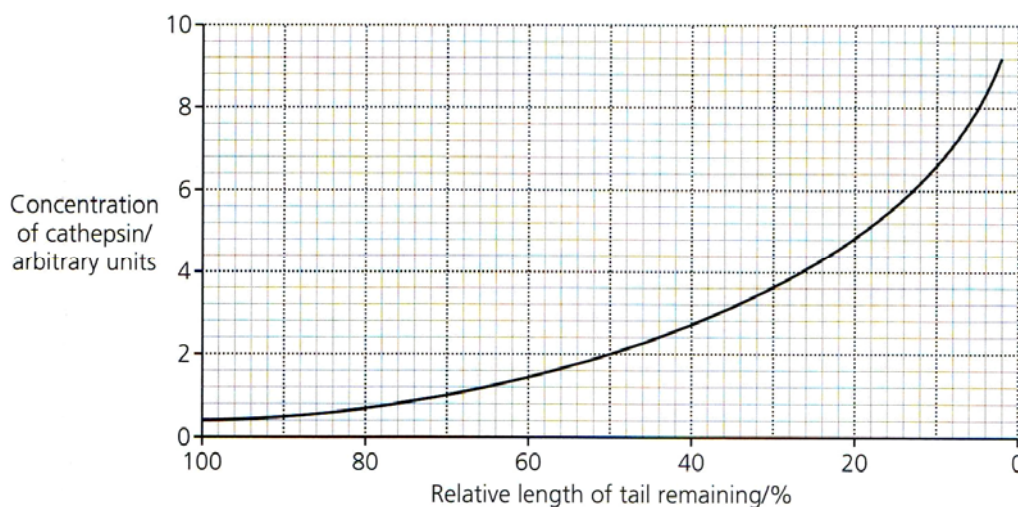


Fig. 1.2

- (i) Suggest two possible causes for the increase in cathepsin concentration in Fig. 1.2.
- Phagocytic cells containing cathepsin e.g. macrophages, increase in number during the regression of the tadpole tail, leading to increased levels of cathepsin;
 - Inactive enzymes (proenzymes) become activated;
 - Lysosomes in cells burst, releasing cathepsin;
 - Any Valid Points; [max 2]
- (ii) Explain an advantage to the developing frog when the proteins in the tadpole's tail break down and the products are absorbed.
- Reabsorbed products of protein breakdown (e.g. amino acids) are used to build up new protein molecules;
 - In limbs and organs etc. of developing tadpole, material from the breakdown of the tail is not wasted;
 - Developing tadpole need not have to acquire all the amino acids from diet; [max 2]
- (iii) Apart from protein-digesting enzymes, suggest one other type of digestive enzyme that you would expect to show a similar change in concentration as the tadpole's tail shortens. Give a reason for your answer.
- RNase (ribonucleases), Dnase (deoxyribonucleases), phospholipases, glycosidases etc. or any valid example;
 - Appropriate reason with reference to the type of molecules present e.g. phospholipases to breakdown structural phospholipids molecules in cell membranes of cells of the tail or Dnases to breakdown DNA in nucleus of unwanted cells etc or any valid point; [max 2]

[Total: 13]

Question 2

Fig.2.1 is a pedigree which traces the inheritance of alkaptonuria, a biochemical disorder. Affected individuals, indicated here by the coloured circles and squares, are unable to break down a substance called alkapton, which colours the urine and stains body tissues.

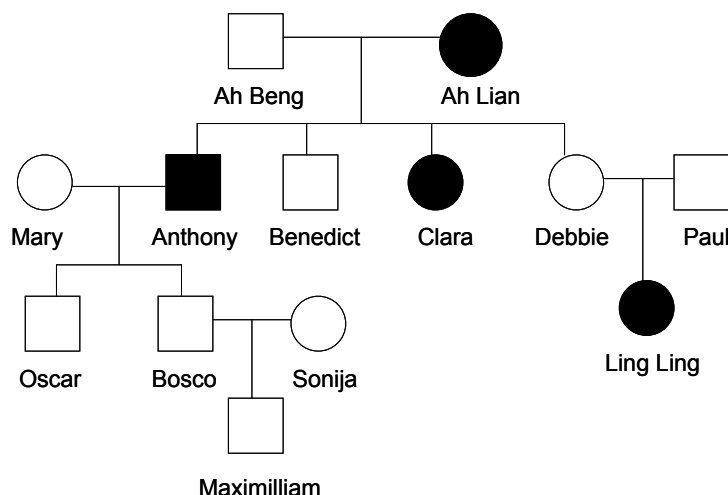


Fig. 2.1

- (a) (i) Is alkaptonuria a dominant or recessive disorder? Explain your answers. [1]
 - **Recessive AND**
 - **Benedict and Debbie are not affected but Lingling is affected / must be – heterozygous for Lingling to be affected (wtte)**
- (ii) Name **two** individuals whose specific genotypes cannot be determined. [2]
 - **Mary, Sonija and Maximilliam (any 2 get 2 mark)**
- (iii)

Oscar's genotype is Aa. If he marries a phenotypically normal female, what is the probability that his child will have alkaptonuria? Explain your answer. [3]

- **prob of phenotypically normal female can be heterozygous /carrier i.e.Aa or homozygous normal i.e. AA OR ½ chance for that to happen**
 - **chances of having affected child with heterozygous /carrier female is ¼ therefore, an affected child with phenotypically normal female is ½ x ¼ = 1/8**

- (b) An inbred variety of maize, variety **A**, with finely striped leaves was found to be resistant to the fungus that causes the disease corn leaf blight.

Plants of variety **A** were crossed with another inbred variety of maize, variety **B**, with entirely green leaves and low resistance to the fungus. All the F₁ generation had entirely green leaves and low resistance.

Using appropriate symbols (indicating what they represent), state the genotype of **A**. [1]

Let G=entirely green leaf and R=low resistance (or any appropriate symbols)
Genotype of A=ggrr

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The above F_1 generation was back-crossed to variety A. Using a genetic diagram, show the expected phenotype of the F_2 generation from this cross. [3]

Parental cross (F_1 gen x variety A) GgRr x ggrr

Gametes GR, gR, Gr, gr gr

1 mark

Punnett square (1 mark)

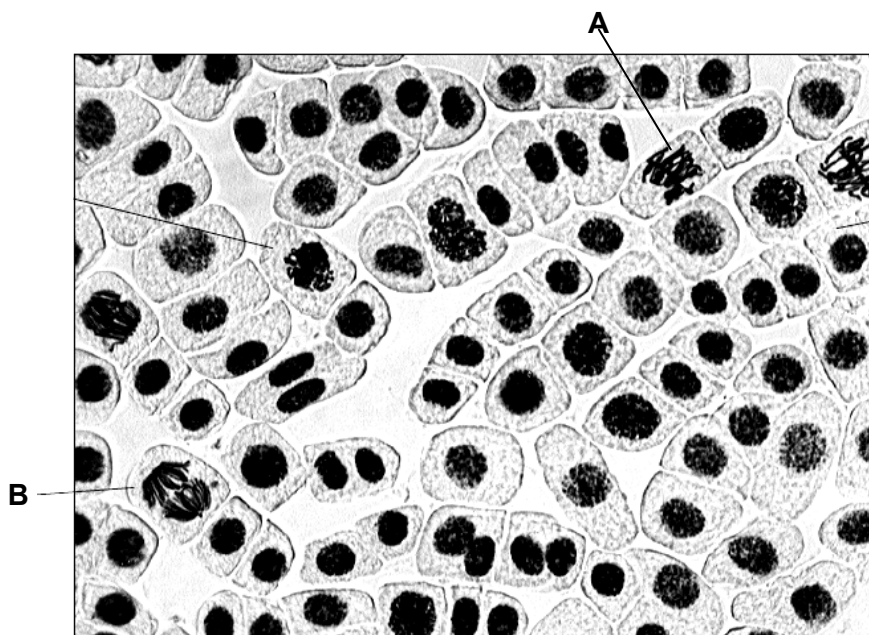
	GR	gR	Gr	gr
gr	GgRr entirely green, low resistance	ggRr striped leaves, low resistance	Ggrr entirely green, resistance	ggrr striped leaves, resistance

Expected 1entirely 1striped 1entirely 1 striped
pheno- green, leaf, low green, leaf,
typic low resistance resis- resistance
ratio resistance: : tance:

[Total: 10]

Question 3

- (a) Fig. 3.1 shows cells from an onion root tip. The root tip has been squashed and stained to show the stages of mitosis.



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Fig. 3.1

(i) Describe what happens to the chromosomes during stage **A**.

- is metaphase
- Chromosomes align along the equator/metaphase plate
- Chromosomes attached to spindle at centromeres

(any 2)

(ii) Explain how stage **B** ensures that the daughter cells formed will be genetically identical.

- Genetically identical sister chromatids
- Centromere divides and chromatids are separated to opposite poles
- Shortening of spindle fibres

(any 2)

(iii) The amount of DNA in cells from the onion root tip tissue was analysed. Some cells were found to have 7.6 units of DNA, others had only 3.8 units. Explain why this is so.

- 7.6 is replicated DNA / cell with 2 sister chromatids joined together at late interphase / prophase / metaphase/ anaphase / early telophase
- 3.8 contains single chromatids / DNA is not replicated / late telophase / early interphase/ before S phase of interphase; non-dividing cells.

OR

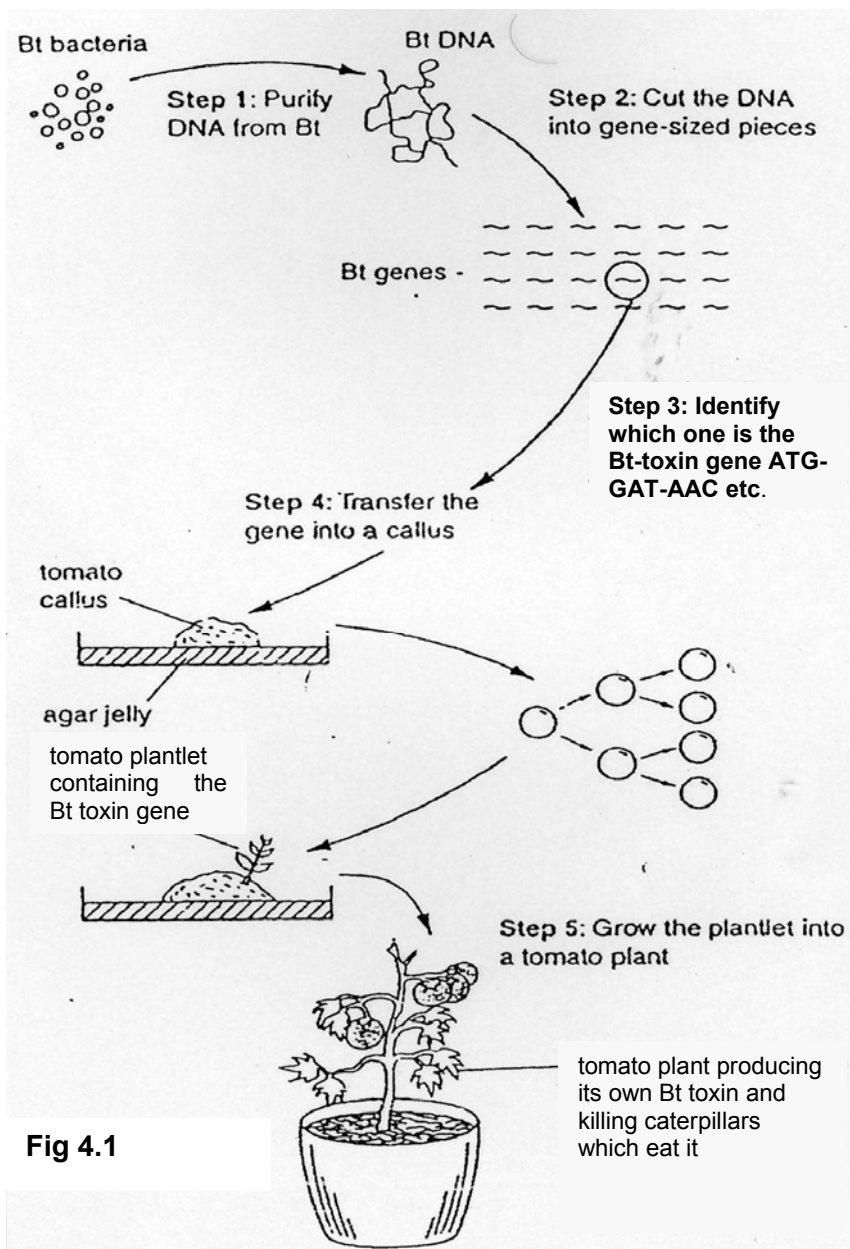
- Polyploidy due to non-disjunction of chromosomes during anaphase
- Results in an extra set of chromosomes in the cell (7.6 units)

[Total: 6]

Question 4

The bacterium *Bacillus thuringiensis* (Bt) makes the Bt-toxin which is toxic to *Lepidopteran* caterpillars and has no effect on most other groups of insects, invertebrates and vertebrates (including Man). The dried bacteria have been used as a biological control measure, sprayed in suspension onto crop plants. However, it becomes ineffective after a few days. The ideal solution would be to have the crop plants make their Bt-toxin. Since Bt-toxin is a protein, this is quite feasible using genetic engineering techniques. Fig.4.1 below shows how tomato plants are genetically engineered to produce their own toxin.

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(a)(i) State why it was considered a better option to engineer crop plants to make the toxin, rather than to spray the plants with the dried bacteria.

- Sprayed insecticide (toxin) may be washed off by rains;
- Underside of leaves must be sprayed to reach insects on undersurfaces of leaves – thorough application is needed is hence time-consuming;
- Dependent on weather conditions for optimal spraying effectiveness;
- Toxin can degrade on exposure to environment e.g. heat, cold etc. – lose effectiveness;
- Costly as require frequent reapplication;

[R: inconvenient, ineffective more laborious without specifying due to short-term effect]

(ii) In **Step 1**, the DNA from *Bt* is extracted and purified. State **two** differences between *Bt* DNA and tomato plant DNA. [2]

Compared to plant DNA, *Bt* DNA is

- Absence of introns / higher % of non-coding regions
- Absence of histones
- Circular DNA
- Genes arranged as operons

[Reject: “fewer genes”, “*Bt* DNA does not carry tomato genes”, “*Bt* DNA is prokaryotic DNA” etc.]

Students must clearly state ways in which the two types of DNA differ.

(iii) Suggest an alternative method of obtaining the T-toxin gene, instead of the method outlined in the diagram. [1]

Use reverse transcriptase to convert mRNA for T-toxin to cDNA sequence

(iv) How could the gene be ‘tagged’ to identify which cells had taken up the gene?

- Use of antibiotic-resistance genes / herbicide-resistance genes;
- Use of fluorescent markers;

(v) State **two** possible hazards of genetic engineering of crop plants.

- Transmission of genes in pollen;
- Genetically modified crops becomes a weed;
- Closely related wild plants become resistant to herbicide / “superweed”;
- Emergence of resistant insects;
- Impact on organic farming;

[R: ethical issues, cos qn is looking at hazards]

- (b) Many crops have been genetically modified (GM) to increase their tolerance to a specific herbicide. Herbicides are chemicals that kill plants.

Britain delayed approving any commercial growing of GM crops until the results of research were available. Research between 2000 and 2003 investigated the effects of GM crops on the environment. Fields were divided and half sown with GM and half with conventional crops. Table shows some of the results of this research.

	Weeds biomass / grams per m ²		Plant-eating insects / mean number per sample		Insect-eating insects / mean number per sample	
	Conventional	GM	Conventional	GM	Conventional	GM
maize	8	14	11.57	15.18	11.09	11.32
beet	22	4	17.14	12.27	15.93	14.73
Oilseed rape	46	13	20.75	16.67	14.15	11.88

Data from *Philosophical Transactions of the Royal Society* 2003

Researchers found different amounts of weeds and insects with different crops.

- (i) With which **conventional** crop did they find the **least** weeds and insects? Justify your answer. [1]

- Maize(½)
- (½) 8g of weeds compared to 22g (beet) and 46g (oilseed rape) AND ~11 insects (plant-eating & insect-eating) per sample compared to ~17 and 20 plant-eating insects (for beet and oilseed rape respectively) / ~15 and 14 insect-eating insects (for beet and oilseed rape respectively) per sample; [accept if simply mention least weed and insect per m² and per sample]

- (ii) Which GM crop appears to be least harmful to weeds and insects **relative to** the same conventional crop? Justify your answer. [2]

- Maize;
- Weeds and insects increased with GM maize by 14g of weeds up from 8g (GM maize) AND 15.18g of plant-eating insects up from 11.57g OR 11.32g insect-eating insects up from 11.02g (conventional maize crops);;
- Weeds and insects decreased with GM beet and GM oilseed rape;

- (iii) Suggest a reason why the insect population fell when herbicide tolerant GM beet was grown.

- Less weeds; hence plant-eating insects die from lack of food;
- Less plant eaters means less food for insect eaters;

[No marks for confusion between herbicide and insecticide i.e. 'insects cannot eat beet']

[Total: 11]

Section BAnswer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

Question 5

- (a) You have found the coding sequence of a gene coding for a growth hormone. Outline one method by which this gene could have been

(i) isolated and [4]

- Restriction enzymes that act on restriction sites flanking the growth hormone gene is used to cut human DNA
- DNA fragments are separated by agarose gel electrophoresis
- The separated DNA fragments are then transferred to a nitrocellulose filter by Southern Blotting. Radioactive gene probe is then used to hybridize with the CFTR gene
- Position of the bound probe is determined by autoradiography
- DNA extracted from the gel
- Gene is amplified by polymerase chain reaction

OR

- Design primers flanking the growth hormone gene
- Carry out PCR using primers
- Use gel electrophoresis to confirm presence of amplified fragment by its relative thickness
- Extract DNA from gel OR use PCR solution

OR

- Isolate total RNA from tissue that is actively expressing the human growth hormone e.g. anterior pituitary gland
- Free oligo-dT serve as primers by hybridising to the 3' poly(A) tail of each mRNA template.
- Reverse transcriptase synthesises a cDNA strand using mature mRNA as a template to produce a hybrid.
- Treat RNA/DNA hybrid with RNase H or alkali to partially break down mRNA leaving single-stranded DNA (ssDNA). Remaining mRNA fragments act as primers for DNA polymerase to synthesize the second cDNA

- (ii) cloned into *E. coli* using a plasmid [8]
for the large scale production of the growth hormone.
(Note: Use a plasmid which carries an ampicillin resistance gene and a tetracycline resistance gene.)

Preparation of DNA and plasmid

- Use a restriction enzyme to cut the flanking region of the growth hormone gene to give a staggered end
- Use the same restriction enzyme to cut the plasmid within a selectable marker (eg *ter*^r)

Formation of rDNA

- Mix cut plasmid and cut DNA fragment together for recombination to occur;
- The staggered end will anneal by complementary base pairing forming H-bond between the complementary base pair;
- DNA ligase is also added to seal the nicks by forming phosphodiester bonds in the sugar backbone;

Transformation of bacteria

- Add recombinant plasmid to *E.coli*/bacteria culture for transformation to take place;
- Brief heat shock and add calcium chloride to facilitate uptake of recombinant DNA. (1/2)

Identification of desired colonies

- Culture bacteria in agar plate containing ampicillin. Transformed *E. coli* will be able to grow on the plate.
- Carry out replica plating onto agar plate containing tetracycline to identify transformant carrying the growth hormone gene.
- Transformants carrying the growth hormone gene will not be able to grow on it due to insertional inactivation.

Large scale production

- Large scale production of Growth hormone using fermenters
- Growth hormone is then isolated and purified from bacteria extract.

- (b) Describe the role of ribonucleic acids in the Central Dogma. [8]

- 1) Central dogma involves the transcription of DNA to RNA and the translation of RNA to protein
- 2) There are 3 types of ribonucleic acids – rRNA, tRNA and mRNA
- 3) mRNA acts as a template for translation (protein synthesis)
- 4) it is a single stranded, small enough to move out of nuclear pores to cytoplasm
- 5) where it attaches to ribosome to start translation
- 6) mRNA contains triplet bases or codon that matches via complementary base pairing with specific anti-codon in tRNA which is attached to specific amino acid determining the sequence of amino acid in the polypeptide
- 7) through the enzymatic reaction of specific aminoacyl tRNA synthetase, specific amino acid is added to tRNA in the process termed amino acid activation
- 8) the tRNA attached to specific amino acid then moves to the ribosome where the anti-codon pairs (via complementary base pairing) with the codon in mRNA ensuring that the correct amino acid is added to the polypeptide chain
- 9) rRNA together with proteins make up the ribosome which is the site of translation
- 10) the ribosomes provide the peptidyl-tRNA, aminoacyl-tRNA and exit sites where the translation of polypeptide occurs
- 11) the bigger subunit of ribosomes also provide the enzyme, peptidyl transferase which catalyses the formation of peptide bond between the amino acid in the polypeptide chain

Question 6

- (a) The products of photosynthesis are used for the formation of storage and structural compounds in plants.

- (i) Describe the process by which these storage and structural compounds are formed. [6]

Starch

- **α -glucose** ;;
- are linked by **α -1,4** ;; and
- **1,6 glycosidic bond at branch points** ;;
- through **Condensation reaction** ;; where water molecule is lost

glycogen

- **β – glucose** ;;
- are linked by **β -1,4 glycosidic bond**;;
- each **alternate glucose is rotated 180°** to each other ;;
- **H-bonds** formed between chains.

- (ii) Discuss the significance of the structure in the functioning of these storage and structural compounds. [7]

Starch - storage

Large molecule makes it **insoluble** hence does **not affect osmotic potential of cell**;;
Helical and branched structure makes it **highly compact for storage**.
Highly branched allows it to be **easily hydrolysed** by enzymes **into glucose**;;
Large number of glucose as monomers – **high energy content** ;;

Glycogen - structural

alternate glucose is rotated 180° to each other results in linear unbranched chain;;
Linear unbranched chain – **impt as structural molecule** ;;
H-crosslinks increases **tensile strength** ;;
Staggered-end – **no major line of weakness** ;;

- b) Discuss the 2 categories of transport proteins found in cell membrane. [7]

transport proteins have hydrophilic domains that allow hydrophilic/charged,polar molecules that cannot transverse the hydrophobic core of the phospholipids bilayer to pass through;
category 1 : channel proteins - are transmembrane proteins;
may be gated, which means they can open to allow the facilitated diffusion of solutes or they can be closed to prohibit diffusion;
this feature allows cells to regulate the movement of solutes. In addition, solutes move through channels from a high to a low solute concentration, down a concentration gradient;
e.g.voltage gated Na^+ channel;
may not be gated form an open passageway for the direct facilitated diffusion of ions or molecules across the membrane;
e.g. aquaporins;
category 2: carrier proteins;
bind the solute on one side of the membrane which results the protein undergoing a conformational change that allows access of the solute to the opposite side of the membrane;
pumps are a type of protein carrier that usually use ATP to move solutes against a concentration gradient (from a low solute concentration to a high solute concentration).
e.g. Na^+ - K^+ pump;
some carriers transport solutes by facilitated diffusion,down a concentration gradient
e.g. glucose transport carriers, Glu T1, across red blood cells